

Illusion of Weight: The Impact of Virtual Dumbbell Size and Vibration on Isometric Exercise in VR

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ABSTRACT

We investigated whether multisensory feedback affects motor control by measuring limb steadiness during an isometric dumbbell exercise in a virtual environment, manipulating virtual dumbbell size (normal vs. large) and tactile feedback (with vs. without vibration). Conditions were counterbalanced per participant. We recorded hand position changes, physiological responses, and subjective feedback from 27 participants aged 18-30. Results showed the largest vertical hand displacement with tactile feedback and significant effects of both dumbbell size and tactile stimuli on physiological responses. These findings suggest that multisensory VR influences limb steadiness, though effect sizes were small and interaction effects indicate further study is needed.

Index Terms: Limb steadiness, Isometric exercise, Human perception

1 INTRODUCTION

Substantial research has revealed that human perception can be easily modified through virtual reality (VR). Multisensory cues, including visuoproprioceptive [1], visuomotor [2], and visuotactile feedback [3], can jointly serve as powerful factors for evoking and manipulating human perceptual experiences. These findings underscore the important role of multisensory feedback in shaping human perception toward illusory experiences, and this technology has been actively adopted in rehabilitation [4] contexts. While VR is being explored to examine muscle function changes in dynamic rehabilitative settings, there is a lack of understanding regarding how artificially induced perceptual stimuli impact certain aspects of motor control, which have yet to receive much attention, such as *limb steadiness*. To address this, we established the following research questions: “Does the multisensory immersive exercise experience affect limb steadiness during an isometric exercise task?” In our isometric exercise experience in VR, we collected data including dominant hand position, physiological signal (GSR, and HR), and a subjective responses for perceived physical exertion, user experiences (presence and engagement), and user preference. The results indicated that visual stimuli and tactile feedback affected the user’s ability to hold the arm position steadily.

2 METHODS

2.1 Study Design

Participants performed an isometric exercise task in which they held a physical dumbbell at shoulder level and were instructed to keep it as still as possible throughout the experiment. To standardize intensity, each participant selected a load corresponding to a moderate

Table 1: Demographics: 27 participants were involved. We collected data on age, gender, height, weight, workout frequency, and VR experience. The mean was followed by the standard deviation in parentheses. Participants’ workout frequency and VR experience with Head Mounted Displays (HMDs) were collected using a 1–5 Likert scale (“Not at all” to “Everyday”)

	Male	Female
# of participant	18	9
Age	21 (3.9)	21 (4.2)
Height (cm)	177.8 (6.86)	162.56 (6.60)
Weight (kg)	83.46 (21.77)	62.14 (14.38)
Workout Frequency	2.14 (1.15)	2.06 (1.16)
VR Experience	1.89 (1.02)	1.11 (.33)

exertion level (RPE 4–5) using the Borg Rating of Perceived Exertion scale [5], determined by incrementally testing heavier weights until the appropriate level was reached. Participants performed 30-second trials for each experimental condition. The study used a 2×2 within-subject design with 27 participants (18 male, 9 female), manipulating virtual dumbbell size (normal vs. large) and tactile feedback (with vs. without vibration) in a VR environment. Across all conditions, we collected hand position data, physiological responses, perceived exertion, and subjective user experience through questionnaires.

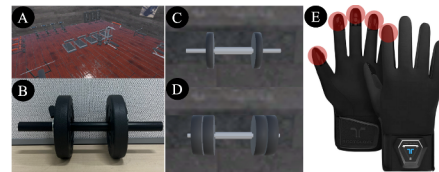


Figure 1: Experiment Apparatus indicates: (a) VR exercise environment, designed to resemble a typical gym. (b) physical dumbbell; its weight is adjusted by adding or subtracting plates based on the participant’s RPE scale. (c) virtual dumbbell, which we manipulate to be a normal size, and (d) large-sized dumbbell. (e) tactile gloves used in our study.

2.2 Environment Design

When the isometric exertion began, the virtual dumbbell was displayed without a virtual avatar body (See Figure 1 (C-D)). For the vibration, we used Bhaptics TactGlove DK1¹. The tactile glove provides vibration feedback through actuators at the finger tips (See Figure 1 (E)). We used a maximum intensity of approximately 0.8G, powered at 3.3V. The tactile feedback remained active throughout the entire duration of each condition. During the task, the bHaptics tactile glove was worn on the dominant hand, and the EmotiBit physiological sensor was placed on the non-dominant hand resting on an armrest (see Figure 2).

2.3 Participants and Procedure

Each participant first selected a dumbbell corresponding to moderate intensity (RPE 4–5). The selected dumbbell was placed in the

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¹<https://www.bhaptics.com/tactsuit/tactglove-dk2/>

participant's dominant hand, and the VR system instructed them to lift and maintain the isometric hold until the "Exercise Over" prompt appeared (see Figure 2). Each condition lasted 30 seconds. After each trial, the HMD and dumbbell were removed, participants reported their RPE, completed a subjective questionnaire, and rested for three minutes (or longer if needed). This procedure was repeated for all four experimental conditions. The entire session lasted approximately 30–45 minutes.

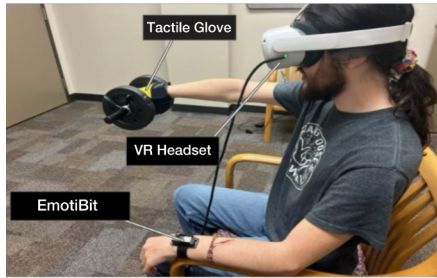


Figure 2: Experiment Setup: Participants performed an isometric exercise task, wearing a tactile glove on their dominant hand.

3 RESULTS

From the reported RPE, we confirmed that a similar moderate perceived intensity (between 4.2 (2.07) - 4.78 (2.09)) was achieved across all conditions. A Two-Way Repeated Measures ANOVA was conducted for the objective measure.

Y-position As shown in 3, there was a significant difference for tactile feedback ($F(1, 72) = 3.81, p = 0.05$) with trivial effect size ($\eta_p^2 < 0.02$). An interaction effect was not found.

Galvanic Skin Response As shown in 4, there were significant differences for virtual dumbbell size ($F(1, 1832) = 16.52, p = 0.01$) with trivial effect size ($\eta_p^2 < 0.01$), and for tactile feedback ($F(1, 1832) = 17.40, p = 0.01$) with trivial effect size ($\eta_p^2 < 0.01$). In addition, we found an interaction effect ($F(1, 1832) = 41.53, p = 0.01$) with trivial effect size ($\eta_p^2 < 0.01$).

Heart rate As shown in 4, there were significant differences for virtual dumbbell size ($F(1, 1928) = 25.842, p = 0.01$) with trivial effect size ($\eta_p^2 < 0.03$), and for tactile feedback ($F(1, 1928) = 4.52, p = 0.03$) with trivial effect size ($\eta_p^2 < 0.01$). In addition, we found an interaction effect ($F(1, 1928) = 8.54, p = 0.01$) with trivial effect size ($\eta_p^2 < 0.01$).

Subjective Responses As shown in 5, we did not find a significant difference for virtual dumbbell size and tactile feedback in Presence and Engagement. Around 70% of participants preferred the conditions with tactile feedback.

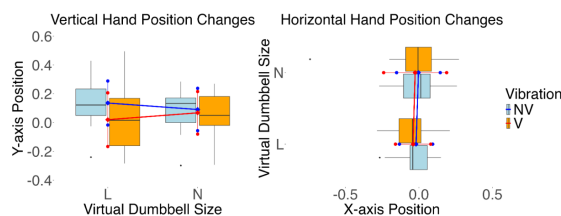


Figure 3: Box plot of vertical and horizontal hand position

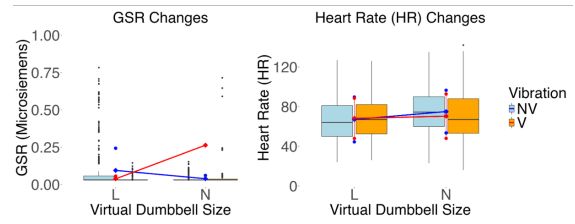


Figure 4: Box plot of physiological changes: GSR and HR

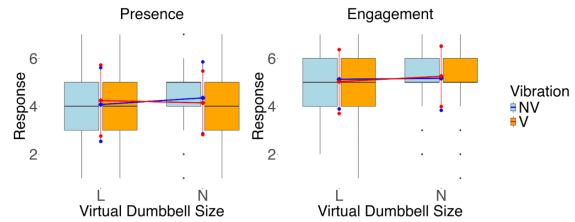


Figure 5: Box plot of Presence and Engagement

4 CONCLUSION

We found a significant difference in vertical hand position movement with tactile feedback, with the large virtual dumbbell combined with vibration showing the highest variability. This suggests that multisensory VR experiences can alter limb stability during isometric exercise, possibly causing perceptual confusion about dumbbell weight and prompting participants to adjust their hand position. Therefore, we focused on the standard deviation of hand displacement rather than the mean, and additionally calculated the vertical hand position difference between the first and last 3 seconds. The Large with No-vibration group showed the largest change ($M = -0.08$), while other conditions exhibited smaller differences, indicating that the visual effect of the large dumbbell impacted limb steadiness. We also observed condition-dependent differences in physiological signals (GSR and HR), suggesting that multisensory feedback modulated sympathetic activation. Vibration appeared to reduce GSR, possibly due to distraction, but this effect was diminished with the large virtual dumbbell, likely due to increased task focus. Overall, multisensory immersive exercise influences limb stability, with tactile feedback as the most salient factor. Although interaction effects were limited, trends in hand movement and significant physiological differences indicate further investigation is warranted.

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